

Work of Fiction: Using Speculative Design to Deliberate on the Future of Hiring

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ABSTRACT

AI recruitment systems are increasingly being deployed to automate the hiring process. This is especially true for companies that hire large numbers of people. AI recruitment software is used in all phases of the process, from receiving and sorting applications, candidate screening, selection process, and communication of results to applicants. While some argue that AI recruitment may remove interviewer bias, others believe there is a danger of dehumanizing the interviewees' experience. In either case, AI recruitment is set to ignite a significant paradigm shift in hiring practices. For this poster, we discuss the results of a participatory speculative design project conducted to engage both HR recruiters and job applicants to understand their concerns. Specifically, we present six scenarios and a diegetic prototype that were used to familiarize participants with the AI recruitment process. We discuss how these served as catalysts for meaningful conversations about an often obfuscated topic.

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1 INTRODUCTION

In this paper, we bring together the domains of speculative design and algorithmic hiring, while raising questions about ethics and assessment of AI systems. We build a future world in a diegetic video that highlights issues of incorporating AI in the hiring process. We screened this video for job applicants and HR professionals and discussed issues and mitigation methods. Our research revolved around (i) What is HR professionals' and job applicants' discernment of current algorithmic hiring practices? (ii) How do they speculate on the future of algorithmic hiring and the perceived implications?

2 RELATED WORK

Design fiction is defined as 'the deliberate use of diegetic prototypes to suspend disbelief about change' and aims to explore and criticize alternative futures by constructing speculative scenarios

told through fictional cultural artefacts [21, 23]. Design fiction narratives are weaved with the help of theoretical and practical future studies methods [8, 14].

A metric has been proposed to evaluate design fiction based on its design and criticality. [3]. The former is done while evaluating design fiction's plausibility, separating design fiction from science fiction, ranging from 'preposterous' to 'preferable' on the 'futures cone' [2]. Strong design fiction can demonstrate a technology's sociopolitical effects, lead to critical dialogue, and provide new perspectives in HCI research [22].

Recruitment is a crucial task for any professional organization. Technology has been amalgamated in this pipeline with the introduction of automation for resume screening, interview analysis and supervision during the pre-interview rounds [16]. With the introduction of AI in hiring systems, companies can efficiently process larger volumes of data efficiently. However, several issues have been raised about this shift concerning both employees and employers. These can be placed into six categories as follows [17]:

Emotional: Some HR professionals are not willing to welcome the change due to a feeling of inadequate control, need for cognitive and emotional trust and dehumanization of the hiring process [5].

Mental: Adapting a new system leads to increased mental load for a candidate, especially considering the black box nature of algorithmic hiring [11].

Bias: The data used to train hiring algorithms can be biased, inaccurate, and unrepresentative, due to which human biases can be reinforced and exacerbated by algorithms [7, 10, 18, 19].

Manipulation: AI is susceptible to easy manipulation with heavy dependency on a computer program written by an external actor. Algorithm transparency has been suggested as a mitigation measure. However, as per the transparency paradox, the more people trust an algorithm, the more it is prone to misuse, posing a stalemate [13].

Privacy: Considering that Artificial Intelligence runs on data and data collection is notoriously high today, the use of data-heavy algorithms raises questions about privacy, pervasiveness [20].

Social: Algorithmic hiring is expected to gamify and introduce artificial values to the hiring process, which may not be the best for a complex social setup like hiring, which demands sincere communication [7].

3 METHODOLOGY

3.1 Building a Diegetic Prototype

We deployed several props to build a world in our diegetic prototype to provoke debate and conduct a philosophical inquiry. The scenario is set in 2030, where AI handles the recruitment and onboarding of individuals in corporate workspaces. We investigate this narrative with the help of two young women applying for a

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position at Technopolis, a gigantic yet mysterious corporation, and their interaction with the company's AI recruiter. We explore this future by speculating artefacts, changing social norms, and other elements of world-building [12].

The origin of this imagined future is based on weak signals emerging in current job hiring and marketplace practices. We provide viewers with an opportunity to reflect on Human-AI interaction, AI explainability, AI pervasiveness, AI ethics, and the future of AI and decision making. We used the future studies technique of Futures Triangle [8] to identify trends leading to historical adoption of technology in workplaces, views on data privacy, and prioritization of values like efficiency. Using Emerging Issue Analysis [8], we gauged the emerging weak signals of AI being used in the workplace and using these trends; we anticipate the future.

The criticality comes from how we change people's perspectives with our depiction of the domain. We change the structure of recruitment interviews, reflecting thought and instigating critique about our current practices. Interviews are supposed to be profoundly personal and give a fair representation of a person's ability in our current age. However, we change this norm by browsing the entire online data available about a person, their genetic characteristics, microexpressions, medical history, EEG, and ECG readings using AI. This is a representation of the pervasiveness of technology. The AI also judges the capability of the person interviewed and decides their fate and line of work, shifting the control and authority from humans to AI. It almost makes humans the slaves of machines, as conceived by Marx with the increasing reliance on technology [4, 24]. This makes us question the functionality of AI and propose ethical and legal changes within our current systems.

3.2 Workshop

Our primary recruitment method was leveraging group posts and direct messages via social media. We utilized a combination of snowball [6] and selective sampling [15], communicating with 10 interested participants (6 female and 4 male) - 7 past/ current job

applicants who had gone through a job recruitment process in the past 2 years and 3 HR professionals who had contributed towards recruiting prospective candidates in the past 2 years. All participants had an Indian origin and were either completing or had completed an undergraduate degree. The ages of the participants ranged from 20 to 48. We fabricated a diegetic prototype for the workshop to engross participants and facilitate critical analysis of AI hiring. The workshop was conducted virtually over a Zoom meeting recorded per the participants' consent. It was initiated with a brief introduction to AI Hiring and Speculative Design to provide participants with better context. This was followed by screening the diegetic prototype. As described in table 1, six case scenarios were displayed to the participants consecutively to encourage discussion about all six categories of issues described in the related work section [17].

The participants were probed through contextual inquiry [9] as they critically reflected on elements of each stage of the hiring pipeline: Identifying and Attracting Candidates, Candidate Screening/ Processing, and Communication and Selection of Candidates while drawing from our prototype to speculate as to what they think the future hiring could look like [16]. All audio-video data from the Zoom meeting and interview notes obtained from the workshop were anonymized and uploaded into a password-protected online repository. The obtained data was then coded using thematic analysis [1], and the consequent findings are discussed further in the next section.

4 FINDINGS AND DISCUSSION

The scenarios that our participants were subject to in the workshop gave birth to a discussion about the advantages and disadvantages of introducing AI in different hiring processes and the design future created in the video Now Hiring, while drawing comparisons with the current real-world situation. Table 1 illustrates the scenarios and the rationale behind them - the issues, as described in the related work section. The information gathered during the workshop and generated insights are summarized here. Participant quotes are added to the insights column.

Issue	Scenario	Insights
Scenario 1 (Privacy, Manipulation)	You are the CEO of Walmart, and you need to hire 5000 store employees by the end of the week. There are over 20000 applications that need to be reviewed. What do you use? AI? Or a Human?	<i>"AI's perspective is black and white on things like confidence or other interpersonal traits. It does not account for the grey area - which is where most humans lie"</i> - Participant_03 In the initial rounds, mass recruitment was pointed out as a job for AI, followed by human lead interviews. AI's judgement on softer skills like confidence was binary, which does not hold for humans. In rejection scenarios, feedback from AI is anticipated to be mechanical, non-descriptive and insensitive.
Scenario 2 (Emotional)	You need to hire a nanny for your child. Your friend had previously hired a nanny and recommended them to you. Alternately, LinkedIn* recommends you a nanny matching the description of the profile that you are looking for. Whom do you hire?	<i>"Every family has their set patterns and expectations about child raising. I'd rather meet both applicants to judge their relationship with my child before taking the final call"</i> - Participant_09 Our participants highlighted the importance of human interaction for sensitive jobs, which they concluded could not be understood from a person's profile, unlike a recommendation, which is a more personal option since someone is vouching for another person. The participants speculated on a future with crowdsourcing feedback on one's LinkedIn profiles to help with emotional hiring decision-making.

Scenario 3 (Social, Bias)	You are a part of your local residential committee, consisting of 9 other members, working on hiring a handyman for the residential complex. The candidates for the job differ in age, race, and gender. Some members of the committee are biased. You have to either get the committee to interview candidates or an AI. Which one do you choose?	<i>"At the end of the day, you want to be satisfied with your decision as a human, so your last deciding factor should be your instinct."</i> - Participant_05 Informants discussed that humans always have rational or orthodox biases. The former is based on experiences with different skill sets, and the latter is based on discrimination against certain groups. AI has biases while judging personal and interpersonal skills based on the limited data it has been trained on. They discussed a hybrid, skill-based approach to eliminate biases. The first elimination, often technical, should be done by AI to eliminate orthodox biases, followed by personal interactions between recruiter and applicant to eliminate rational biases.
Scenario 4 (Mental, Manipulation)	You are hiring a news correspondent from a pool of candidates who belong to the PWD (people with disability) community. The participants have diverse physical and mental abilities. You need to sift through all resumes. Do you do it yourself, or do you seek the help of AI?	<i>"There is a gap between the human and their resume, which exists regardless of AI vs human sifting. Thus, for the resume sifting part of the process, use AI to save time and human resources."</i> - Participant_01 This scenario pushed the conversation towards training AI models to make data representative of all categories of people, including people with disabilities, since AI may directly reject them if there is no representation in the training data set, to make a diverse workforce.
Scenario 5 (Emotional, Privacy)	You are an entrepreneur with an environmental startup and need to recruit a CEO to manage its day-to-day functioning. You receive applications from people from different industries than yours. Do you use an AI to make hiring decisions or choose the CEO yourself?	<i>"If I am not familiar with a domain, I'll hire consultants and then make the decision, with or without AI's help, based on new knowledge I've gained"</i> - Participant_10 The idea of hiring consultants was dropped here to hire for fields new to the recruiter, rather than depending on, or giving authority to AI, since understanding the rationale behind decisions will be harder for both sides. An alternate idea for the same was to depend on AI to create the balance between lack of skill knowledge on the recruiter's part, especially in smaller organizations that lack the resources to hire expensive consultants.
Scenario 6 (Bias, Manipulation, Social)	Your company needs you to hire a new graduate for the role of Assistant Software Developer and a veteran for the position of the Chief Technical Officer. However, due to prior commitments, you only have time to interview for one of the positions, with the other being interviewed by an AI recruiter. Where would you involve AI?	<i>"A CTO can alter the company's ego, and a fresher can be molded into whatever the role requires - there isn't much thought required to decide where AI will be involved."</i> - Participant_06 From a company's point of view, hiring someone at an executive position can alter a company's ego and perspective, have legal implications and is a higher cost for the company. Recruiters would want minimal chances of error, thereby giving humans the authority to make decisions for this post. Participants articulated that for entry-level jobs, they do not have much experience and are usually moldable for the company's benefit. In cases like this, AI can conduct technical tests and test a person's learning capability and potential.

5 CONCLUSIONS

We have explored the future of hiring through future studies and speculative design, a field previously unexplored. By presenting this work in the workshop, we gather that interviewers and applicants alike prefer to leave all hiring decisions for higher position jobs to humans. The introduction of AI is more welcome for mass hiring and starter jobs. Previous work in critical design fiction has not considered the entirety of issues that occur during recruitment. We contributed to the design futures domain by deploying world-building elements in our diegetic prototype, such as speculating artefacts, ecology, behavior and interactions. We have attempted to do an exhaustive study as a part of this project. AI decisions are highly based on the training dataset, which needs to be representative. It is suggested that fairness must be taken into account at every step while building a recruitment algorithm, that is Identifying and

Attracting Candidates, Candidate Screening/ Processing, and Communication and Selection of Candidates, and HR professionals must carefully and frequently review the results to ensure the algorithm's performance. Using bias suppression techniques like adding a set of constraints on the classification objective to achieve results like parity in hiring from different groups will successfully make algorithms robust. This study also opens avenues for further research and conversation in HCI around structuring stringent guidelines for algorithmic recruiting.

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